AI-Based Battery Lifecycle Assessment and Second-Life Management Using Digital Battery Passports

Week 3: Data cleaning and health indicators

Mon 16 Nov - Fri 20 Nov 2026 · Sprint 3 · Overall progress 30%

Student name		**Roll no.**	
Faculty in charge		**Date submitted**	Fri 20 Nov 2026

On track **This week in one line:** A one-sided Hampel filter removed 120 interrupted cycles while keeping real capacity regeneration; charge-curve health indicators correlate up to r = 1.00 with SOH.

1. Goal for the week

Remove measurement artefacts and build health indicators that a battery management system can observe during ordinary charging.

2. Day-by-day work

Day	Work done
Mon 16 Nov	Analysed the outliers: interrupted discharges (drops) versus capacity regeneration after rest (real, jumps up).
Tue 17 Nov	Implemented a one-sided Hampel filter (11-cycle window, 3 x MAD, two passes) that removes only drops.
Wed 18 Nov	Two-sided spike repair for charge features; 9-cycle median SOH for end-of-life detection.
Thu 19 Nov	Built health indicators: CC charge time, CV charge time, 3.9-4.2 V rise time, resistance, cycle count; per-cell relative versions.
Fri 20 Nov	Correlation analysis, figure, 7 new tests (12 total), this report.

3. Completed deliverables

- **batterylife/features/cleaning.py** and **indicators.py**.
- **data/processed/features.csv**.
- tests/test_features.py (7 tests).

4. Results

Health indicator	Correlation with SOH, NASA	Correlation with SOH, CALCE
CC charge time	0.94	1.00
3.9-4.2 V rise time	0.93	0.97
CV charge time	-0.90	-0.73
Resistance	-0.56	-0.77
Outliers removed	0 of 636 (regeneration kept)	120 of 3,995

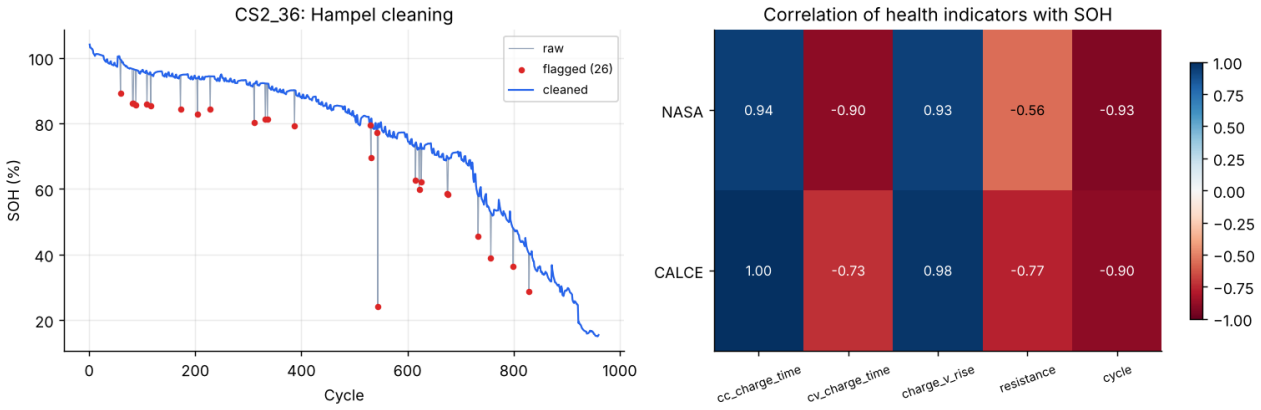


Figure 3. Left: cleaning on CS2_36 (red points removed). Right: correlation of each indicator with SOH.

5. Problems faced and how they were solved

Problem	Resolution
CALCE cells charge from empty, so CC charge time is almost a direct capacity measurement.	Added a 'partial-charge' feature set without it, tested next week.
CS2_38 resistance sensor shows a long plateau (cycles 30-140).	Documented as a sensor artefact; robust baseline limits its effect.

6. Plan for next week

- Train and compare SOH models with leave-one-cell-out validation.
- Add calibrated confidence intervals.

Faculty remarks

Faculty signature: _____ Student signature: _____